

Lecture 2: Data Models

An Introduction to the Relational Model

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What is a Data Model?

- A data model is a collection of concepts for describing data.
- It provides a way to describe the design of a database at the physical, logical, and view levels.
- It is a set of tools for describing:
 - Data
 - Data relationships
 - Data semantics
 - Data constraints

Categories of Data Models

- **Record-Based Data Models:**

- Data is organized in fixed-format records of several types.
- Examples: Relational Model, Network Model, Hierarchical Model.

- **Object-Based Data Models:**

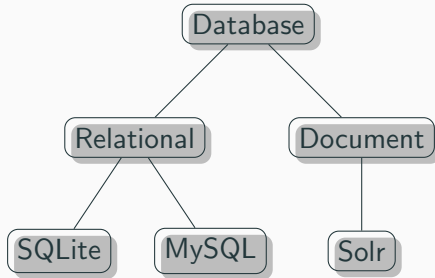
- Use concepts such as entities, attributes, and relationships.
- Example: Entity-Relationship (ER) Model.

- **Physical Data Models:**

- Describe data at the lowest level (how it is stored on disk).

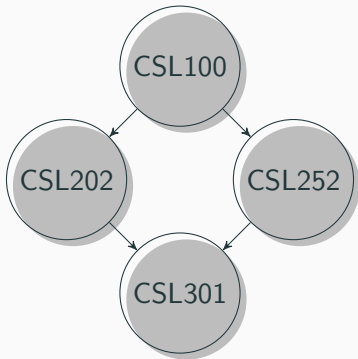
Hierarchical Model

- Data is organized into a tree-like structure.
- The data is stored as records which are connected to one another through links.
- A record is a collection of fields, with each field containing only one value.
- A child record can only have one parent.



Network Model

- Similar to the hierarchical model, but a child record can have multiple parents.
- This allows for more complex relationships to be modeled.
- Data is organized in a graph structure.



The Relational Model

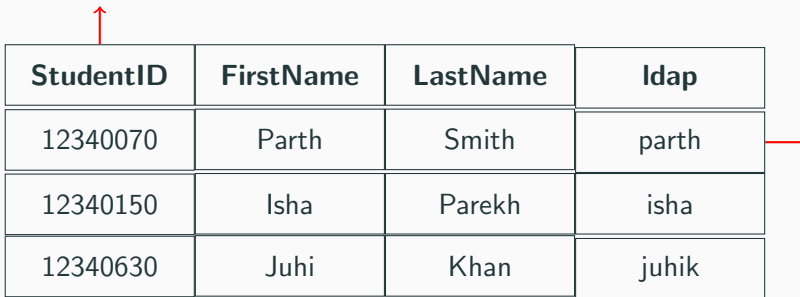
- Proposed by E.F. Codd in 1970.
- The most widely used data model.
- Data is organized into **tables**, which are called **relations**.
- A relation is a set of **tuples** (rows), where each tuple is a set of **attribute** (column) values.

Example Relation: Students

StudentID	FirstName	LastName	Idap
12340070	Parth	Smith	parth
12340150	Isha	Parekh	isha
12340630	Juhi	Khan	juhik

Relational Model Terminology

Column/Attribute/Field



StudentID	FirstName	LastName	Idap
12340070	Parth	Smith	parth
12340150	Isha	Parekh	isha
12340630	Juhi	Khan	juhik

Row/Tuple/Record

More Terminology

- **Domain:** The set of allowed values for each attribute.
 - Example: The domain of 'StudentID' might be the set of all 3-digit integers.
- **Degree:** The number of attributes in a relation.
 - The 'Students' relation has a degree of 4.
- **Cardinality:** The number of tuples in a relation.
 - The 'Students' relation has a cardinality of 3.

Schema vs. Instance

- **Schema:** The logical design of the database. It specifies the name of the relation and the name and type of each attribute.
 - Example: 'Students(StudentID: integer, FirstName: string, LastName: string, Idap: string)'
- **Instance:** The actual data in the database at a particular point in time.
 - The table we saw earlier is an instance of the 'Students' relation.

Keys in the Relational Model

A key is a way to uniquely identify tuples in a relation.

- **Superkey:** An attribute, or a set of attributes, that uniquely identifies a tuple.
- **Candidate Key:** A minimal superkey (no proper subset is a superkey).
- **Primary Key:** A candidate key that is chosen by the database designer as the principal means of identifying tuples.

Example of Keys

StudentID	FirstName	LastName	Idap
12340070	Parth	Smith	parth
12340150	Isha	Parekh	isha
12340630	Juhi	Khan	juhik

Superkey

StudentID

FirstName

LastName

Idap

Example of Keys

StudentID	FirstName	LastName	Idap
12340070	Parth	Smith	parth
12340150	Isha	Parekh	isha
12340630	Juhi	Khan	juhik

Candidate Key

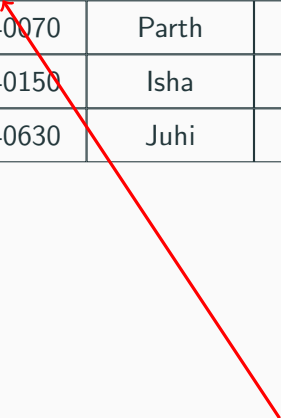
StudentID

Idap

Example of Keys

StudentID	FirstName	LastName	Idap
12340070	Parth	Smith	parth
12340150	Isha	Parekh	isha
12340630	Juhi	Khan	juhik

Primary Key
Choice



StudentID

- A **foreign key** is a key used to link two tables together.
- It is a field (or collection of fields) in one table that refers to the primary key in another table.
- This is how we create relationships between relations.

Foreign Key Example

Students

<u>StudentID</u>	Name
12340070	Parth
12340150	Isha

Enrollment

<u>CourseID</u>	StudentID
CSL300	12340070
CSL300	12340150
CSL312	12340070



Thank You!!

Questions?